# Artificial Intelligence in BioInformatics

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## Short-bio

- Professor Dr. Do Phuc is currently working at the University of Information Technology, VNU-HCM. His research areas include bioinformatics, text processing, artificial intelligence, machine learning, deep learning, natural language processing, knowledge graphs, question answering, verification, and big data processing.
- The professor has participated in numerous research projects across various fields, including bioinformatics, text processing, natural language processing, text understanding systems, and reasoning over knowledge graphs. He has published more than 60 scientific papers in prestigious international and national journals. He is the author of 8 reference books and textbooks in the fields of bioinformatics, computer science, and applications.
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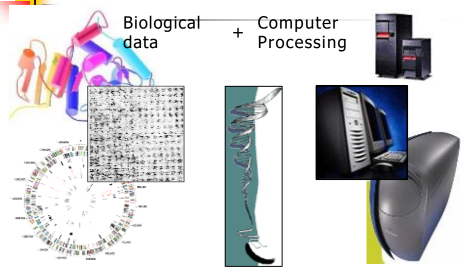
## Outline

- Introduction to Bioinformatics:
  - Bioinformatics
  - Objectives of Bioinformatics
  - Biological Information
  - Protein Databases and Structures
  - Development of Bioinformatics Research
  - Informatics Techniques in Bioinformatics
  - Research Topics in Bioinformatics
- Artificial Intelligence in Bioinformatics
  - Artificial Intelligence
  - Video Clip: AI in Bioinformatics
  - Applications of AI in Bioinformatics
- Information on the Research and Teaching of Bioinformatics by the Presenter



## Bioinformatics



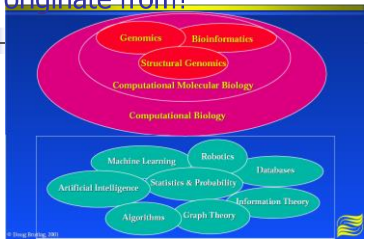



## Informatics?

- Derived from the French word "informatique," informatics is the science of processing information.
- Definition: Informatics (informatics) is the science of processing information.
- Related Fields:
  - - Medical informatics
  - - Bioinformatics
  - - Applied informatics in management and business



## Where does bioinformatics originate from?




The data from the Human Genome Project has driven the development of the field of bioinformatics.

# BioInformatics?

- (Molecular) Bioinformatics
- Idea and Definition: Bioinformatics understands biology at the molecular level (in a physico-chemical sense) and uses informatics techniques to comprehend and organize biological information on a large scale.

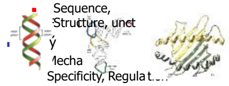


# Molecular biology as an information science

- The main foundation of molecular biology

```
DNA
 -> RNA
  -> Protein
   -> Phenotype
    -> DNA
```

- Phân tử
  - Sequence, Structure, unct
- ý
  Mecha
  Specificity, Regula t...



•Genetic material

•Information transfer (mRNA)
•Protein synthesis (tRNA/mRNA)
•Some catalytic activity

- Central strategy of bioInformatics

```
Genomic Sequence Information
 -> mRNA (level)
  -> Protein Sequence
   -> Protein Structure
    -> Protein Function
     -> Phenotype
```

- Large Amounts of Information
  - Standardized
  - Statistical

•Most cellular functions are performed or facilitated by proteins.
•Primary biocatalyst
•Cofactor transport/storage
•Mechanical motion/support
•Immune protection
•Control of growth/differentiation

(idea from D Brutlag, Stanford, graphics from S Strobel)

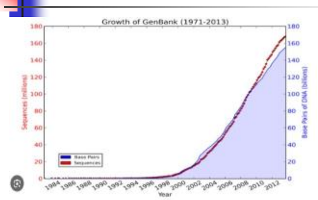


# The growth of biological data

- GenBank has become a crucial database in biological research and has grown rapidly in recent years, doubling its size approximately every 18 months. Version 250.0, released in June 2022, contains over 17 trillion nucleotide bases from more than 2.45 billion sequences.
- The Protein Data Bank (PDB) holds 214,111 PDB entries and includes 14,482 protein structures as of 2023.
- The primary goal of the Human Genome Project (HGP), conducted from 1990 to 2003, was to determine both the DNA sequences and "the location of about 100,000 genes in the human genome."
- Additionally, numerous genome research projects identifying protein structures encoded by genes have generated a vast amount of biological information, which continues to become increasingly diverse and abundant.



# Large-scale information: The growth of GenBank




Popularity of Bioinformatics depicted by growth of GenBank (source:... | Download Scientific Diagram

| GenBank Data | | |
|---|---|---|
| Year | Base Pairs | Sequences |
| 1982 | 680,338 | 606 |
| 1983 | 2,274,029 | 2,427 |
| 1984 | 3,368,765 | 4,175 |
| 1985 | 5,204,420 | 5,700 |
| 1986 | 9,615,371 | 9,978 |
| 1987 | 15,514,776 | 14,584 |
| 1988 | 23,800,000 | 20,579 |
| 1989 | 34,762,585 | 28,791 |
| 1990 | 49,179,285 | 39,533 |
| 1991 | 71,947,426 | 55,627 |
| 1992 | 101,008,486 | 78,608 |
| 1993 | 157,152,442 | 143,492 |
| 1994 | 217,102,462 | 215,273 |
| 1995 | 384,939,485 | 555,694 |
| 1996 | 651,972,984 | 1,021,211 |
| 1997 | 1,160,300,687 | 1,765,847 |
| 1998 | 2,008,761,784 | 2,837,897 |
| 1999 | 3,841,163,011 | 4,864,570 |
| 2000 | 11,101,066,288 | 10,106,023 |
| 2001 | 15,849,921,438 | 14,976,310 |
| 2002 | 28,507,990,166 | 22,318,883 |
| 2003 | 36,553,368,485 | 30,968,418 |
| 2004 | 44,575,745,176 | 40,604,319 |
| 2005 | 56,037,734,462 | 52,016,762 |



# Definition of Bioinformatics

- The rapid growth of biological data, computational tools have become indispensable for analyzing and processing biological data.
- IT can manage vast data sources, analyze diverse and constantly evolving data in the natural world.
- Bioinformatics is considered an interdisciplinary research field that integrates computing techniques and information organization using computer devices with common techniques and tools in molecular biology.
- The integration of two scientific disciplines, IT and biology, has driven strong research into biological technology, especially physiological studies at the genetic level.
- Rapid advances in computer techniques in data acquisition enable easy retrieval of biological sequence data.



# Video clip: What is bioinformatics?

## The objectives of BioInformatics

- Organizing data for managing and accessing information.
- Developing tools and resources to support the analysis of biological data, such as comparing specific protein sequences with well-characterized functional sequences.
- Using these tools to analyze data and interpret results meaningfully in biology.



## The objectives of BioInformatics

- Traditional biological studies often assess individual systems by comparing them with related individuals.
- In the field of bioinformatics, it is possible to manage analyzed biological data globally via the Internet to actively support comparative processes.



## Integrated information with molecular biology

- The information resources are divided into biological DNA sequences, biological protein sequences, molecular structures, gene sequences, and genomes.
- Raw DNA sequences are strings composed of 4 basic nucleotide characters (A, C, G, T). On average, each gene has a length of about 1,000 base pairs.



## Integrated information with molecular biology

- GenBank: Release 250.0, published in June 2022, contained over 17 trillion nucleotide bases in more than 2,45 billion sequences.
- Hiện có khoảng 300,000 trình tự protein đã biết, một protein của vi khuẩn có chiều dài khoảng 300 codon.
- Dữ liệu cấu trúc đại phân tử là một dạng phức tạp của thông tin.
- CSDL PDB (Protein Database) có 214,111 PDB entries và có 14,482 protein structure,2023



## Molecular biological information DNA

DNA sequences
Encoded or not?
Analyzed into genes?
4 bases: AGCT
~1 K in a gene, ~2 M in genome
~3 Gb Human

```
atggcaattaaaattggtatcaatggttttggtcgtatcggccgtatcgtattccgtgca
gcacaacaccgtgatgacattgaagttgtaggtattaacgacttaatcgacgttgaatac
atggcttatatgttgaaatatgattcaactcacggtcgtttcgacggcactgttgaagtg
aaagatggtaacttagtggttaatggtaaaactatccgtgtaactgcagaacgtgatcca
gcaaacttaaactggggtgcaatcggtgttgatatcgctgttgaagcgactggtttattc
ttaactgatgaaactgctcgtaaacatatcactgcaggcgcaaaaaaagttgtattaact
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tcacaaaacatcattccatcttcaacaggtgcagcgaaagcagtaggtaaagtattacct
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gttgatttaacagttaatcttgaaaaaccagcttcttatgatgcaatcaaacaagcaatc
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gaagatgctgttgtttctactgacttcaacggttgtgctttaacttctgtatttgatgca
gacgctggtatcgcattaactgattctttcgttaaattggtatc . . .

. . . caaaaatagggttaatatgaatctcgatctccattttgttcatcgtattcaa
caacaagccaaaactcgtacaaatatgaccgcacttcgctataaagaacacggcttgtgg
cgagatatctcttggaaaaactttcaagagcaactcaatcaactttctcgagcattgctt
gctcacaatattgacgtacaagataaaatcgccatttttgcccataatatggaacgttgg
gttgttcatgaaactttcggtatcaaagatggtttaatgaccactgttcacgcaacgact
acaatcgttgacattgcgaccttacaaattcgagcaatcacagtgcctatttacgcaacc
aatacagcccagcaagcagaatttatcctaaatcacgccgatgtaaaaattctcttcgtc
ggcgatcaagagcaatacgatcaaacattggaaattgctcatcattgtccaaaattacaa
aaaattgtagcaatgaaatccaccattcaattacaacaagatcctctttcttgcacttgg
```



## Molecular biological information: Protein

There are 20 letters (20 amino acids):

- ACDEFGHIKLMNPQRSTVWY but not BJOUXZ.

- Protein sequences have an average length of 200 to 300 amino acids.
- The number of known protein sequences is approximately 200,000.

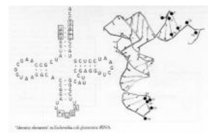
```
d1dhfa_   LNCIVAVSQNMGIGKNGDLPWPPLRNEFRYFQRMTTTSSVEGKQ-NLVIMGKKTWFSI
d8dfr_____LNSIVAVCQNMGIGKDGNLPWPPLRNEYKYFQRMTSTSHVEGKQ-NAVIMGKKTWFSI
d4dfra_   ISLIAALAVDRVIGMENAMPWN-LPADLAWFKRNTL--------NKPVIMGRHTWESI
d3dfr___TAFLWAQDRDGLIGKDGHLPWH-LPDDLHYFRAQTV--------GKIMVVGRRTYESF

d1dhfa_   LNCIVAVSQNMGIGKNGDLPWPPLRNEFRYFQRMTTTSSVEGKQ-NLVIMGKKTWFSI
d8dfr_____LNSIVAVCQNMGIGKDGNLPWPPLRNEYKYFQRMTSTSHVEGKQ-NAVIMGKKTWFSI
d4dfra_   ISLIAALAVDRVIGMENAMPW-NLPADLAWFKRNTLD--------KPVIMGRHTWESI
d3dfr___TAFLWAQDRNGLIGKDGHLPW-HLPDDLHYFRAQTVG--------KIMVVGRRTYESF

d1dhfa_   VPEKNRPLKGRINLVLSRELKEPPQGAHFLSRSLDDALKLTEQPELANKVDMVWIVGGSSVYKEAMNHP
d8dfr_____VPEKNRPLKDRINIVLSRELKEAPKGAHYLSKSLDDALALLDSPELKSKVDMVWIVGGTAVYKAAMEKP
d4dfra_   ---G-RPLPGRKNIILS-SQPGTDDRV-TWVKSVDEAIAACGDVP------EIMVIGGGRVYEQFLPKA
d3dfr___---PKRPLPERTNVVLTHQEDYQAQGA-VVVHDVAAVFAYAKQHLDQ----ELVIAGGAQIFTAFKDDV

d1dhfa_   -PEKNRPLKGRINLVLSRELKEPPQGAHFLSRSLDDALKLTEQPELANKVDMVWIVGGSSVYKEAMNHP
d8dfr_____-PEKNRPLKDRINIVLSRELKEAPKGAHYLSKSLDDALALLDSPELKSKVDMVWIVGGTAVYKAAMEKP
d4dfra_  -G---RPLPGRKNIILSSSQPGTDDRV-TWVKSVDEAIAACGDVPE-----  IMVIGGGRVYEQFLPKA
d3dfr___-P--KRPLPERTNVVLTHQEDYQAQGA-VVVHDVAAVFAYAKQHLD----QELVIAGGAQIFTAFKDDV
```

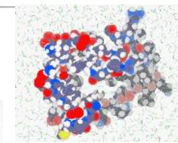
# 20 amino acids

Here are 20 amino acids along with some of their corresponding codons in DNA:
Alanine (Ala): GCU, GCC, GCA, GCG
Arginine (Arg): CGU, CGC, CGA, CGG, AGA, AGG
Asparagine (Asn): AAU, AAC
Aspartic Acid (Asp): GAU, GAC
Cysteine (Cys): UGU, UGC
Glutamine (Gln): CAA, CAG
Glutamic Acid (Glu): GAA, GAG
Glycine (Gly): GGU, GGC, GGA, GGG
Histidine (His): CAU, CAC
Isoleucine (Ile): AUU, AUC, AUA
Leucine (Leu): UUA, UUG, CUU, CUC, CUA, CUG
Lysine (Lys): AAA, AAG
Methionine (Met): AUG (also serves as the start codon)
Phenylalanine (Phe): UUU, UUC
Proline (Pro): CCU, CCC, CCA, CCG
Serine (Ser): UCU, UCC, UCA, UCG, AGU, AGC
Threonine (Thr): ACU, ACC, ACA, ACG
Tryptophan (Trp): UGG
Tyrosine (Tyr): UAU, UAC
Valine (Val): GUU, GUC, GUA, GUG



# Thông tin Sinh học phân tử
# Cấu trúc đại phân tử sinh học



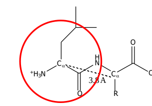




# Molecular biological information
# Protein structure

- Statistics on Number of XYZ triplets
  - 200 residues/domain -> 200 CA atoms, separated by 3.8 A
  - Avg. Residue is Leu: 4 backbone atoms + 4 sidechain atoms, 150 cubic A or a bead of diameter 6.6 A
    - => ~1500 xyz triplets (=8x200) per protein domain
  - 10 K known domains but only ~300 folds

```
ATOM      1  C   ACE     0       9.401  30.166  60.595  1.00 49.88      1GKY  67
ATOM      2  O   ACE     0      10.432  30.832  60.722  1.00 50.35      1GKY  68
ATOM      3  CH3 ACE     0       8.876  29.767  59.226  1.00 50.04      1GKY  69
ATOM      4  N   SER     1       8.753  29.755  61.685  1.00 49.13      1GKY  70
ATOM      5  CA  SER     1       9.242  30.200  62.974  1.00 46.62      1GKY  71
ATOM      6  C   SER     1      10.453  29.500  63.579  1.00 41.99      1GKY  72
ATOM      7  O   SER     1      10.593  29.607  64.814  1.00 43.24      1GKY  73
ATOM      8  CB  SER     1       8.052  30.189  63.974  1.00 53.00      1GKY  74
ATOM      9  OG  SER     1       7.294  31.409  63.930  1.00 57.79      1GKY  75
ATOM     10  N   ARG     2      11.360  28.819  62.827  1.00 36.48      1GKY  76
ATOM     11  CA  ARG     2      12.548  28.316  63.532  1.00 30.20      1GKY  77
ATOM     12  C   ARG     2      13.502  29.501  63.500  1.00 25.54      1GKY  78
. . .
ATOM   1444  CB  LYS   186      13.836  22.263  57.567  1.00 55.06      1GKY1510
ATOM   1445  CG  LYS   186      12.422  22.452  58.180  1.00 53.45      1GKY1511
ATOM   1446  CD  LYS   186      11.531  21.198  58.185  1.00 49.88      1GKY1512
ATOM   1447  CE  LYS   186      11.452  20.402  56.860  1.00 48.15      1GKY1513
ATOM   1448  NZ  LYS   186      10.735  21.104  55.811  1.00 48.41      1GKY1514
ATOM   1449  OXT LYS   186      16.887  23.841  56.647  1.00 62.94      1GKY1515
TER    1450      LYS   186                                              1GKY1516
```





# Genomes

The genome is the total genetic information or all the genes that a specific organism (such as a cell, an individual, or a species) contains. It includes all the genes and the non-coding regions between them. Genes are the basic units of heredity, and the genome contains all the necessary information to build and maintain an organism.

In the context of humans, the genome is referred to as the "human genome." The human genome contains genes that control hereditary traits such as skin color, eye color, height, as well as genes that govern the functions of proteins and many other biological aspects. Research on the genome is a crucial field in genetics and molecular biology to understand the structure and function of genes.



# Molecular biological information
# Genomes

- Integrated data in 1995,
- H. influenzae (bacteria): 1.6 Mb and 1600 genes completed by 1997,
- Yeast: 13 Mb and approximately 6000 genes for yeast in 1998,
- C. elegans (roundworm): ~100 Mb with 19 thousand genes in 1999.
- Over 30 genome maps completed by 2003!
- Human: 3 Gb and 100 thousand genes...



# Human genome project (HGP)

- The Human Genome Project (HGP) is a large-scale international project that began in the early 1990s and officially completed in 2003. The main goal of this project was to sequence the entire human genome, which means identifying the exact sequence of nucleotide pairs in human DNA.
- Thanks to rapid advancements in biological technology and the decreasing cost of genome sequencing, the HGP produced a detailed genome map, providing a clearer understanding of the human genetic structure. This project has provided crucial information for genetic research, medicine, and clinical applications, ranging from disease diagnosis to the development of personalized treatment protocols.
- The results of the HGP have had a profound impact on many scientific and medical fields, opening up new opportunities for research and development in genetics and molecular medicine.

## Human Genome Project

- 1. **Number of Genes:** It is estimated that humans have about 20,000 - 25,000 genes. Before the Human Genome Project (HGP), initial estimates suggested around 100,000 genes, but this number significantly decreased after completion.
- 2. Number of Nucleotides (bp - base pairs):** The human DNA sequence consists of about 3 billion base pairs (bp).
- 3. Single Nucleotide Polymorphism (SNP):** Single nucleotide polymorphisms (SNPs), which are single nucleotide variations in the human genome, occur frequently. The Human Genome Project identified and characterized approximately 10 million SNPs across the entire genome.
- 4. Genome Size:** The genome sequence of each human varies slightly in length but typically ranges from 2.85 billion to 3.2 billion base pairs.
- 5. Time and Cost:** The Human Genome Project spanned from 1990 to 2003 and had an estimated cost of around 3 billion US dollars.



## The key persons of the HGP

- **Francis Collins:** As a molecular biologist, Francis Collins played a pivotal role in leading the project and serving as the director of the National Institutes of Health (NIH).
- **Craig Venter:**A renowned geneticist, Craig Venter made significant contributions to expanding and speeding up the process of genome sequencing.
- **James D. Watson:** The co-discoverer of the DNA double helix structure with Francis Crick in 1953. In the early stages of the HGP, he played a crucial role in promoting the project and providing important insights.
- **Alec Jeffreys:** a British geneticist known for inventing the technique of DNA fingerprinting.



## NCBI Databases



## Video clip: Giới thiệu về NCBI




## NCBI

- The NCBI (National Center for Biotechnology Information) is an organization within the U.S. National Institutes of Health (NIH) responsible primarily for providing and maintaining a range of critical databases in the fields of molecular biology and medicine. Some key databases managed by NCBI include:
- 1. PubMed: This database contains information on scientific articles from various fields of medicine and biology.
- 2. GenBank: GenBank is a large database containing information on nucleotide sequences of DNA, RNA, and proteins collected from various sources.
- 3. BLAST (Basic Local Alignment Search Tool): It provides tools to search for nucleotide and protein sequences in databases to identify similarities and evolutionary relationships.



## NCBI

- 4. **Entrez Gene**: Gene data and related information, including gene location, protein products, and other details.
- 5. **PubMed Central (PMC)**: A free archive of biomedical and life sciences journal literature, providing open access to research documents for the research community.
- 6. **Genome Database (GDB)**: Stores and provides genetic information related to various species.
- These databases support the research community in accessing and sharing crucial information about genes, proteins, and other aspects of molecular biology.

## Protein Database

- The protein sequence databases are classified into primary, mixed, and secondary categories.
- Primary databases contain over 300,000 protein sequences along with their functions, serving as raw data repositories.
- Some commonly used primary databases include SWISS-PROT and PIR, which store protein sequences, functions, structures, and post-translational modifications.
- Mixed databases such as OWL and NRDB compile and curate sequence data.



## Protein Database

- The primary database serves to aggregate and refine data from individual databases. It includes protein sequence data translated from coding regions in DNA sequences.
- Secondary databases consist of information inferred from protein sequences, aiding users in determining whether newly identified sequences belong to known protein families.



## CSDL trình tự protein

- Một trong những CSDL phổ biến là PROSITE, đây là một CSDL chứa các mẫu trình tự ngắn và hồ sơ tổng lược (profile) nhằm biểu thị các vị trí (site) có ý nghĩa sinh học trong protein.
- CSDL PRINTS [27] mở rộng khái niệm này và cung cấp bản tóm tắt dấu vân protein – nhóm của đoạn lặp được bảo tồn để đặc trưng cho họ protein. Đoạn lặp thường cách nhau trong trình tự protein, nhưng vẫn liên tục trong không gian 3D khi protein bị gấp nếp. Bằng việc sử dụng nhiều đoạn lặp, có thể mã hoá các nếp gấp protein, các chức năng trong PROSITE .
- CSDL Pfam [28] chứa các phương án gióng thẳng cột nhiều trình tự (multiple alignment) và hồ sơ  tổng lược Markov ẩn của nhiều protein phổ biến.
- CSDL Pfam-A chứa các phương án gióng cột trong khi Pfam-B là kết quả gom cụm tự động của toàn bộ dữ liệu CSDL SWISS-POT.



## Protein Structure Database

- The Protein Data Bank (PDB) provides 3D structures of biomolecules such as proteins, RNA, DNA, and other complexes. As of August 2000, PDB contained approximately 13,000 structures resolved using X-ray crystallography and NMR methods.
- PDBsum offers individual web pages for each structure within PDB, containing detailed structural analyses, diagrams, and data on interactions between different molecules.



## Protein Structure Database

- The three main databases that classify proteins by structure to identify structural relationships and evolution are CATH, SCOP, and FSSP. These databases employ hierarchical architectures to group proteins based on similarity levels. They include a vast array of unique macromolecules.
- These databases include:
- - Nucleic Acids Database (NDB), focusing on structures related to nucleic acids.
- - HIV Protease Database (HIV Protease), containing structures of HIV-1, HIV-2, and SIV proteases and their complexes.
- - ReLIBase, which houses receptor-ligand complexes.



## Developing research in bioinformatics

- Two main tasks of the field of Bioinformatics are organizing and understanding biological data – the advancement of bioinformatics technology enables expanding biological analyses in both depth and breadth.
- In terms of depth, this includes studies aimed at understanding more about proteins. Starting with a gene, identifying the protein sequence, and predicting the protein structure.
- Through geometric computations, one can predict the shape and surface of proteins, and model molecular interactions. Identifying bonds and inferring protein functions. In practice, these intermediate steps are still challenging to execute accurately, necessitating integration with other methods to achieve desired outcomes.

## Developing research in bioinformatics

- In terms of breadth, it involves methods for comparing genes with other genes, and proteins with other proteins. Initially, simple algorithms were used to compare sequences and structures of related protein pairs. As biological data proliferates, there is a growing need to enhance algorithms for efficient alignment of multiple sequences, extracting sequence motifs or structural motifs to identify protein families, and constructing phylogenetic trees to study protein evolution.
- Ultimately, due to the vast amount of information stored in large databases, the task of comparison becomes more complex, demanding improvements in database organization and management mechanisms.



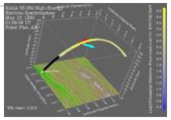
## General computational techniques in bioinformatics

- Database
  - Building, Querying
  - Object DB
- Sequence comparison
  - Text Search
  - 1D Alignment
  - Significance Statistics
  - Google, grep
- Pattern Search
  - AI / Machine Learning
  - Clustering
  - Datamining
- Geometry
  - Robotics
  - Graphics (Surfaces, Volumes)
  - Comparison and 3D Matching (Vision, recognition)
- Physics simulation
  - Newtonian Mechanics
  - Electrostatics
  - Numerical Algorithms
  - Simulation

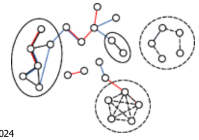


## BioInformatics: new road for computing science

Physics
- Prediction based on physical principles
- Example: accurately determining the orbit of a comet
- Emphasis: Supercomputers, CPUs



- ◊ Biology:
- ◊ Classifying information and discovering unexpected relationships
- ◊ Example: Gene network representation
- ◊ Emphasis: Networks, "up-to-date" databases





## Bioinformatics Topics -- Genome Sequence

- Finding Genes in the DNA sequence
- Genes
- - Introns
- - Exons
- - Promoters
- Repeating features within the genome
- - Statistics
- - Patterns
- - Gene duplications
- Large-scale genomic alignment



## The research topics of BioInformatics

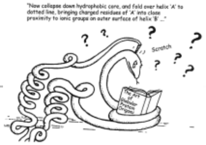
- sequence alignment
  - non-exact string matching, gaps
  - How to align two strings optimally via Dynamic Programming
  - Local vs Global Alignment
  - Suboptimal Alignment
  - Hashing to increase speed (BLAST, FASTA)
  - Amino acid substitution scoring matrices
- multi sequence alignment and consensus Patterns
  - How to align more than one sequence and then fuse the result in a consensus representation
  - Transitive Comparisons
  - HMMs, Profiles
  - Motifs
- Scoring schemes and Matching statistics
  - How to tell if a given alignment or match is statistically significant
  - A P-value (or an e-value)?
  - Score Distributions (extreme val. dist.)
  - Low Complexity Sequences
- Evolutionary Issues
  - Rates of mutation and change



## sequene/structure

- Secondary Structure "Prediction"
  - via Propensities
  - Neural Networks, Genetic Alg.
  - Simple Statistics
  - TM-helix finding
  - Assessing Secondary Structure Prediction
- Structure Prediction: Protein v RNA

"Now collapse down hydrophobic core, and fold over helix 'A' to dotted line, bringing charged residues of 'A' into close proximity to ionic groups on outer surface of helix 'B' ..."




Reproduced in U. Tollemar, "*Protein Engineering i USA*", Sveriges Tekniska Attachéer., 1988

- Tertiary Structure Prediction
  - Fold Recognition
  - Threading
  - Ab initio
- Function Prediction
  - Active site identification
- Relation of Sequence Similarity to Structural Similarity

## The topics of genomes

- Expression Analysis
  - Time Courses clustering
  - Measuring differences
  - Identifying Regulatory Regions
- Large scale cross referencing of information
- Function Classification and Orthologs
- The Genomic vs. Single-molecule Perspective
- Genome Comparisons
  - Ortholog Families, pathways
  - Large-scale censuses
  - Frequent Words Analysis
  - Genome Annotation
  - Trees from Genomes
  - Identification of interacting proteins
- Structural Genomics
  - Folds in Genomes, shared & common folds
  - Bulk Structure Prediction
- Genome Trees



## Ứng dụng kỹ thuật tin học

- Nhiều lĩnh vực Sinh Tin học đòi hỏi các kỹ thuật tin học khác nhau: đối với tổ chức dữ liệu, CSDL sinh học những sử dụng các tập tin phẳng đơn giản.
- Tuy nhiên khi gia tăng số lượng thông tin, các CSDL quan hệ với giao diện Web sẽ ngày càng phổ biến, những kỹ thuật mới cần phát triển bao gồm phương pháp so sánh chuỗi, thuật giải chỉnh thẳng cột, nhận diện đoạn lặp (motif) và các phương pháp học máy, phân cụm và kỹ thuật khai thác dữ liệu. Việc phân tích cấu trúc 3D bao gồm tính toán hình học Euclid kết hợp với ứng dụng cơ bản của hóa lý, thể hiện đồ họa của bề mặt và hình khối, và sự so sánh cấu trúc và phương pháp hợp 3D.



## Applications of Computer Techniques

- In molecular modeling, Newton's mechanism, quantitative mechanism, and molecular mechanism, electrostatic static calculations have been applied.
- In many of these fields, computational methods must be combined with statistical analysis to provide meaningful data and results.



## Research on Protein Structure

- The principles of structured classification of DNA-binding proteins are reflected in SCOP and CATH, first proposed by Harrison and regularly updated to incorporate new structures as they are resolved.
- The classification involves a two-tiered system: the first level groups proteins into 8 folds sharing overall structural features involved in DNA binding, and the second level comprises 54 families of proteins that share structural similarities.



## Research on Protein Structure

- The assembly of a simplified system facilitates the comparison of various combined methods, highlighting the diversity of proteins—complex geometric DNA structures found in nature, but also emphasizing the crucial points of the interactions between DNA helices and main grooves, the primary method of combining over half of the protein families.
- While the number of structures represented in the PDB does not fundamentally reflect the essential interrelationship of different proteins in the cell, it is clear that the helix-turn-helix motif, zinc-coordinating, and leucine zipper motifs are repeatedly used.



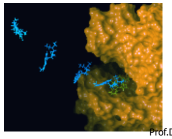
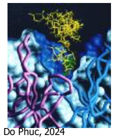
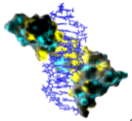
## Research on genomes

- The study of genomes. With readily available biochemical data, genomic studies in bioinformatics focus on organism models and systematic regulatory analysis.
- The consistency of copying factors in the genome depends on the research strategy for recognizing similarities.

## Application: Drug Design

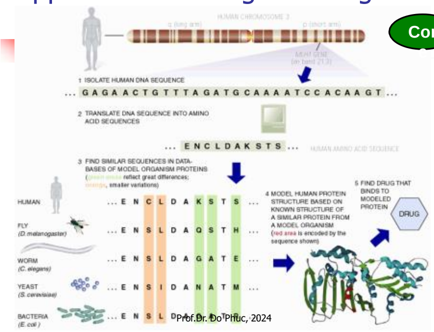
- Understanding How Structures Bind Other Molecules (Function)
- Designing Inhibitors
- Docking, Structure Modeling

(From left to right, figures adapted from Olsen Group Docking Page at Scripps, Dyson NMR Group Web page at Scripps, and from Computational Chemistry Page at Cornell Theory Center).



## Application: Finding Homologs






## Application Finding Homologues

- Find Similar Ones in Different Organisms
- Human vs. Mouse vs. Yeast
  - Easier to do Expts. on latter!

(Section from NCBI Disease Genes Database Reproduced Below.)

Best Sequence Similarity Matches to Date Between Positionally Cloned Human Genes and S. cerevisiae Proteins

| Human Disease | MIM | Human Gene | GenBank Acc# for Human cDNA | BLASTX P-value | Yeast Gene | GenBank Acc# for Yeast cDNA | Yeast Gene Description |
|---|---|---|---|---|---|---|---|
| Hereditary Non-polyposis Colon Cancer | 120436 | MSH2 | U03911 | 9.2e-261 | MSH2 | M84170 | DNA repair protein |
| Hereditary Non-polyposis Colon Cancer | 120436 | MLH1 | U07418 | 6.3e-196 | MLH1 | U07187 | DNA repair protein |
| Cystic Fibrosis | 219700 | CFTR | M28668 | 1.3e-167 | YCF1 | L35237 | Metal resistance protein |
| Wilson Disease | 277900 | WND | U11700 | 5.9e-161 | CCC2 | L36317 | Probable copper transporter |
| Glycerol Kinase Deficiency | 307030 | GK | L13943 | 1.8e-128 | GUT1 | X69049 | Glycerol kinase |
| Bloom Syndrome | 210900 | BLM | U39817 | 2.6e-119 | SGS1 | U22341 | Helicase |
| Adrenoleukodystrophy, X-linked | 300100 | ALD | Z21876 | 2.4e-107 | PXA1 | U17065 | Peroxisomal ABC transporter |
| Ataxia Telangiectasia | 208900 | ATM | U26455 | 2.8e-90 | TEL1 | U31331 | PI3 kinase |
| Amyotrophic Lateral Sclerosis | 105400 | SOD1 | K00065 | 2.0e-58 | SOD1 | J03279 | Superoxide dismutase |
| Myotonic Dystrophy | 160900 | DM | L19268 | 5.4e-53 | YPK1 | M21307 | Serine/threonine protein kinase |
| Lowe Syndrome | 309000 | OCRL | M88162 | 1.2e-47 | YIL002C | Z47047 | Putative IPP-5-phosphatase |
| Neurofibromatosis, Type 1 | 162200 | NF1 | M89914 | 2.0e-46 | IRA2 | M33779 | Inhibitory regulator protein |
| Choroideremia | 303100 | CHM | X78121 | 2.1e-42 | GDI1 | S69371 | GDP dissociation inhibitor |
| Diastrophic Dysplasia | 222600 | DTD | U14528 | 7.2e-38 | SUL1 | X82013 | Sulfate permease |
| Lissencephaly | 247200 | LIS1 | L13385 | 1.7e-34 | MET30 | L26505 | Methionine metabolism |
| Thomsen Disease | 160800 | CLC1 | Z25884 | 7.9e-31 | GEF1 | Z23117 | Voltage-gated chloride channel |
| Wilms Tumor | 194070 | WT1 | X51630 | 1.1e-20 | FZF1 | X67787 | Sulphite resistance protein |
| Achondroplasia | 100800 | FGFR3 | M58051 | 2.0e-18 | IPL1 | U07163 | Serine/threoinine protein kinase |
| Menkes Syndrome | 309400 | MNK | X69208 | 2.1e-17 | CCC2 | L36317 | Probable copper transporter |



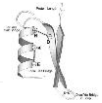
## Application : Finding Homologues (cont.)

- Cross-Referencing, one thing to another thing
- Sequence Comparison and Scoring
- Analogous Problems for Structure Comparison
- Comparison has two parts:
  (1) Optimally **Aligning** 2 entities to get a Comparison **Score**
  (2) Assessing **Significance** of this score in a given **Context**

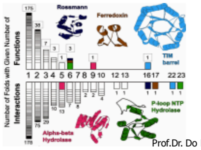
- **Integrated Presentation**
  - Align Sequences
  - Align Structures
  - Score in a Uniform Framework



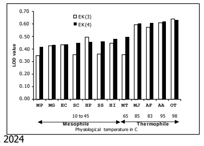
## Data Mining? Global Analysis proteins



**EX-1**: Occurrence of functions per fold & interactions per fold over all genomes




**EX-2**: Occurrence of 1-4 salt bridges in genomes of thermophiles v mesophiles





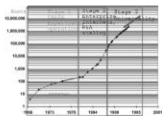

## Video: Applications of the BioInformatics

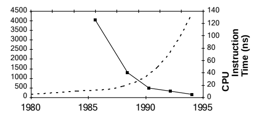
## Large Scale Information: The exponential growth of data requires powerful computing systems to process.

- CPU vs Disk & Net
  - As important as the increase in computer speed has been, the ability to store large amounts of information on computers is even more crucial
- Driving Force in Bioinformatics

(Internet picture adapted from D Brutlag, Stanford)

Internet Hosts



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## Computational Infrastructure for Bioinformatics Analysis

- NGS => Super Big data => super performance computing
- High performance computing infrastructure :
  - Computing cluster :
    - Multiple nodes(servers) with multiple cores
    - High computer memory (64 GB)
    - High performance storage(TB, PB level)
    - Fast networks(10Gb ethernet, infiniband)
  - Operating System : Unix based (Linux)
  - Server room
  - System admin

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## Big Data Processing:Distributing computing

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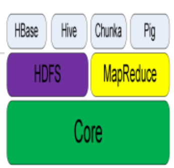
## Giới thiệu Hadoop

- Hadoop là một framework nguồn mở viết bằng Java dùng để chạy những ứng dụng trên 1 cluster lớn được xây dựng trên những phần cứng thông thường.
- Hadoop hiện thực mô hình Map/Reduce chia nhỏ ra thành nhiều phân đoạn, chạy song song trên nhiều node khác nhau.
- Cho phép các ứng dụng có thể làm việc với hàng ngàn node khác nhau và hàng petabyte dữ liệu.
- Hadoop cho phép phát triển các ứng dụng phân tán bằng java,scala, python và một số ngôn ngữ lập trình khác
- Hadoop giải pháp cho SME để xử lý big data

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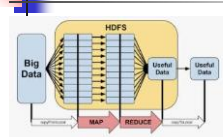
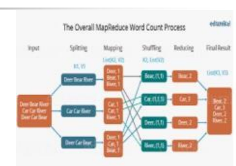
## Map Reduce




Mapreduce Example in Apache Hadoop ...

**MapReduce: Data distribution, distributed computing**

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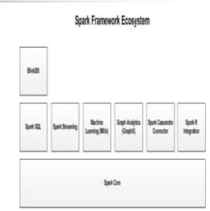
## Apache Spark: Big data processing in the memory

- Fast cluster computing system, compatible with Apache Hadoop
- Works with any distributed storage system supported by Hadoop (HDFS, S3, Avro, ...)
- Improves efficiency through: In-memory computing principles
- MLlib, GraphX, SQL, Stream libraries

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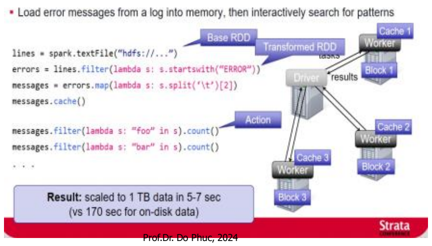
## Spark cluster

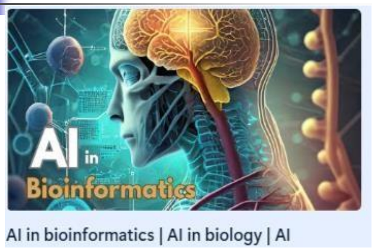





## Trí tuệ nhân tạo là gì?
## AI- Artificial Intelligence



## Ai in BioInformatics






## Overview of AI

- Artificial intelligence (AI) encompasses methods for simulating human intelligence, including activities such as learning, perception, and decision-making.
- When discussing AI, terms such as cognitive science, machine intelligence, machine learning, and deep learning can be used.




## Overview of AI

hese solutions combine various technologies that interact with services such as:

Machine learning - the capability of software to learn from data analysis and create models to perform a task.

Data science - the collection, processing, and analysis of large amounts of data.

Internet of Things (IoT) - involves diverse connected devices used to collect and transmit data.




## Why we use AI in bioinformatics

1.Processing large and complex data: Biological data is often very large and complex, including information from various sources such as genomics, proteomics, metabolomics, and medical imaging. AI has the capability to efficiently process large and complex data, helping to extract important information from chaotic data and simultaneously uncovering complex relationships between different factors.

2.Prediction and classification: Machine learning algorithms can be used to predict the outcomes of biological experiments, such as predicting protein structure, classifying diseases, or genetic traits.

3.Discovering new knowledge: AI has the capability to uncover complex relationships in data, aiding biological researchers in discovering new knowledge about the connections between genes, proteins, and other biological processes..

## Why we use AI in bioinformatics

- 4. Feature-based search: AI can assist in feature-based searches of data, helping to identify important genes, proteins, or biological structures.
- Enhancement for genomic data analysis and decoding: In the field of genomics, AI can aid in decoding gene sequences and identifying gene expressions, helping researchers better understand genetic rules and the impact of genes on health.
- Optimization of the research process: AI can help optimize the research process, from selecting experiments to analyzing data, reducing time and costs.
- In summary, using AI in bioinformatics brings robustness in processing and understanding biological data while providing powerful tools for research and applying biological knowledge to medicine and other fields.



## Expert Systems



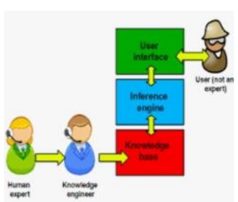
## What is expert system?

- An expert system is a type of knowledge-based system designed for a specific application domain.
- For example, an expert system for credit evaluation in banking. Expert systems operate like a true expert.




## What is expert system?

- The knowledge base is where the system stores the representation of the knowledge it manages, serving as the foundation for the system's activities. The knowledge base includes facts and rules.
- Inference engine: This is the process in an expert system that allows matching the facts in the working memory with the knowledge in the knowledge base to draw conclusions about the problems being solved.






## Machine learning



## Machine Learning

- Machine learning is a field of AI that involves the study and development of techniques that allow systems to learn from data.

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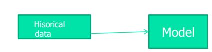
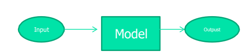


## Học giám sát
## Supervised learning

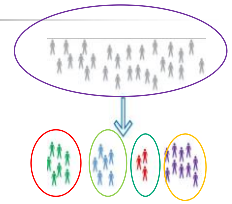
- Dữ liệu học một tập hợp các cặp đầu vào - đầu ra đã được biết trước. (dữ liệu học)
- Hệ thống học sẽ phải tìm cách thay đổi các tham số bên trong của mình để tạo nên một ánh xạ có khả năng ánh xạ các đầu vào thành các đầu ra mong muốn
- Học để nhận diện khách hàng, nhận diện cổ phiếu tiềm năng, học để phát hiện gian lận tài chính,tư vấn, ra quyết định

"open" - The price as of market open.
"close" - The price as of market close.
"high" - The current day's high price.
"low" - The current day's low price.
"volume" - The current day's trading volume.
Source Nasdad, Dự báo chuỗi thời gian



## Unsupervised learning

- The method aims to find a model that fits the observations.
- It does not require a training phase.
- It clusters data based on similarity.



Gom cụm khách hàng. Chiến lược tiếp thị kinh doanh từng cụm khách hàng



## Reinforcement Learning

- Data $xx$ is often not pre-generated but created during the process of an agent interacting with the environment. At each time $tt$, the agent performs an action and the environment generates an observation with an immediate cost , according to some dynamic process. The objective is a strategy for selecting actions to minimize a cost.
- Problems typically solved by reinforcement learning include robot control, games, self-driving cars, and more.

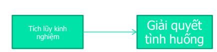



## Artificial neural network



## Artificial neural network

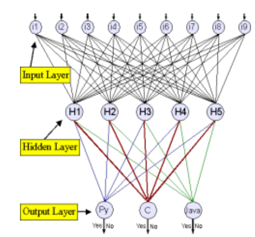
- Artificial Neural Network (ANN) simulates functions of the human nervous system with numerous neurons interconnected to communicate through a network.
- Similar to humans, ANN learns from experience (through learning data), stores those experiences, and uses them in appropriate situations.






## Muttilayer neural network

Input layer: receives inputs for the network.
Output layer: provides outputs for the network.
Hidden layer: This layer, situated between the input and output layers, represents the network's logical inference process.





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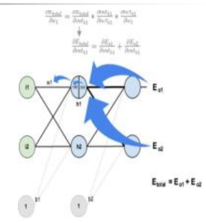
## Training of ANN

- Backpropagation algorithm:
- Provides the training dataset.
- Training criterion: Minimize the output error as much as possible.
- Methodology: Adjust weights of the output layer based on the output, then adjust weights of the hidden layer based on calculations from the output layer.



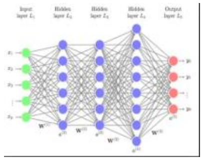


## Deep Learning



## Deep learning

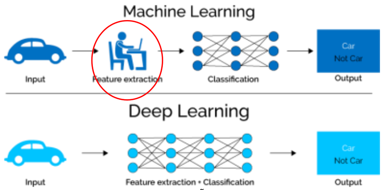
- Deep learning is a branch of machine learning that involves a set of algorithms attempting to model high-level abstractions of data through multiple layers of complex processing.
- Training data is very large, marking a revolution in data analytics.





## Why use deep learning? It allows systems to explore the features of data.

A field of machine learning involves data representation learning, feature extraction. The learning algorithm attempts to understand (at multiple levels) representations using a hierarchical system consisting of multiple layers. Object recognition




## Video Clip:AI in bioinformatics



## Application of AI in bioinformatics

- **Genomics Analysis:**
- Gene and Protein Prediction: Machine learning models can be used to predict the positions of genes and related proteins based on genomic sequence data.
- Comparing Genomes: AI can assist in comparing and analyzing the similarity and differences between genomic sequences to gain a better understanding of evolution and genetic expressions.

- **Predictive Structured Biology:**
- Protein Structure Prediction: AI can assist in predicting protein structures, elucidating their functions and interactions.
- Modeling Network Dynamics: Using machine learning models to simulate the complex network of biological reactions and protein dynamics.

- **Drug Discovery and Development**
- Prediction Model of Drug-Protein Interactions: Applying AI to predict interactions between drug molecules and proteins, speeding up the process of developing new drugs.
- Searching Drug-Genome Relationships: Identifying relationships between gene expression and drug responses to optimize treatment.

- **Biomedical Image Analysis:**
- Segmentation and Recognition of Medical Images: Using machine learning to classify and segment structures in medical images, aiding in diagnosis and disease treatment.
- Prediction of Disease Progression: Utilizing medical images to predict disease progression, facilitating early detection and disease management.

- **Single-Cell Data Analysis:**
- Cell Classification: AI can assist in classifying and understanding gene expression of individual cells, enhancing understanding of cellular diversity and variations within specific tissues.

- **Disease Risk Prediction:**
- Cancer Risk Prediction: Using machine learning models to predict the risk of developing cancer based on genomic and individual medical data.

- **Biomedical Data Management:**
- Handling and Analyzing Big Data: AI can assist in processing and analyzing large volumes of biomedical data, thereby generating valuable insights for research.
- .

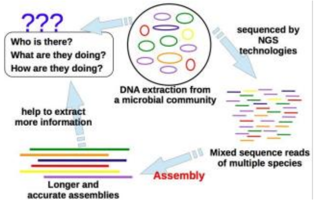


# Metagenomic Analysis

- Metagenomic Analysis is a research method in the field of molecular biology that does not require isolating and culturing microorganisms in the research environment. Instead, it focuses on studying and analyzing DNA or RNA directly extracted from a complex environmental sample, such as soil, water, or the human gut.
- This method often utilizes DNA sequencing techniques to read and analyze gene segments from environmental samples.
- The applications of Metagenomic Analysis are diverse, ranging from environmental ecological studies to analyzing bacteria in the human gut.

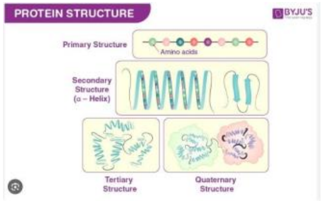


# Metagenomic Analysis






# Protein Structure



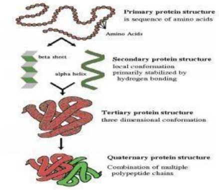



# Secondary Structure Prediction

- Secondary Structure Prediction is the process of predicting the secondary structure of a polypeptide chain (such as a protein sequence) from its amino acid sequence. In proteins, secondary structure is typically divided into three main types: α-helix, β-sheet, and turns.
- Prediction methods for secondary structure often use information from the amino acid sequence and statistical data from known protein databases. Machine learning algorithms, such as Support Vector Machines (SVM) or Neural Networks, are commonly used to train prediction models based on features of the amino acid sequence.
- The protein's secondary structure is crucial for understanding its function because it affects how the protein folds and interacts with other components in its environment. Secondary Structure Prediction can support research into protein function and protein design in the field of molecular biology.



# Secondary Structure Prediction




| Kelas DSSP | Simbol 8 Kelas | Simbol 3 Kelas | Nama Kelas |
|---|---|---|---|
| α-helix<br>$3_{10}$-helix | H<br>G | H | *Helix* |
| B-Strand<br>B-bridge | E<br>B | E | *Sheet* |
| Loop/Irregular<br>B-Turn<br>Bend<br>π-helix | L<br>T<br>S<br>I | C | *Coil / Loop* |

## Secondary Structure Prediction

**Sekuens 1: Panjang sekuens = 67**

| | |
|---|---|
| **SPP** | VPSLATISLENSWSGLSKQIQLAQGNNGIFRTPIVLVDNKGNRVQITNV TSKVVTSNIQLLLNTRNI |
| **SSP 3** | CCCHHHHHHHHHHHHHHHHHHHHHHCCCCCEEEEEEEEECCCCCCCCEEE EECCCHHHHHCECCECCHHHC |
| **SSP 8** | LLLHHHHHHHHHHHHHHHHHHHHHHHTTTTTEEEEEEEEELLSSSSLEEEEET TSHHHHHTBLLBLLGGGL |
| **PSSP 3** | CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC CCCCCCCCCCCCCCCCCC |
| **PSSP 8** | LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL LLLLL |



## Secondary Structure Prediction

- Machine learning methods :
  - Multi Layer Perceptron
  - Support Vector Machine
  - Convolutional Neural Network
  - Recurrent Neural Network (as a sequence prediction)



## DNA2Vec

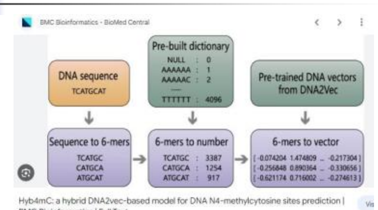



## DNA2Vec

- One common representation of long DNA sequences is to divide them into shorter k-mer components. Unfortunately, directly encoding k-mers as one-hot vectors is susceptible to the curse of dimensionality. Moreover, the distance between any pair of one-hot vectors is equal, which becomes problematic when applying state-of-the-art machine learning algorithms to address issues in biological sequence analysis.
- An article proposes a new method for training distributed representations of variable-length k-mers. Our method is based on the popular word2Vec embedding model, trained on a two-layer neural network. Our experiments provide evidence that the synthesis of dna2Vec vectors is equivalent to concatenating nucleotides. We also demonstrate a correlation between Needleman-Wunsch similarity scores and cosine similarity scores of dna2Vec vectors.



## DNA2Vec with k-mer






## DNA sequences->DNA2Vec

```
# Define the DNA sequences
dna_sequences = [
    'ATCGATCGATCG',
    'CGATCGATCGAT',
    'ATCGATCGCGCG',
    'CGCGCGCGATCG'
]
```

```
DNA2Vec Embeddings:
[[ 2.5048446e-02 -1.1486478e-02  1.2936710e-02  8.5565262e-06
  -1.3162974e-02  1.8269813e-03 -5.9240721e-03 -2.2172282e-02
  -3.3170097e-03  1.6112803e-02]
 [ 2.5048446e-02 -1.1486479e-02  1.2936711e-02  8.5565262e-06
  -1.3162974e-02  1.8269815e-03 -5.9240735e-03 -2.2172282e-02
  -3.3170099e-03  1.6112803e-02]
 [ 2.2364942e-02 -1.4732639e-02  1.8300207e-02 -8.3380518e-04
  -1.5008114e-02 -1.5104640e-03 -1.3945134e-03 -2.6418010e-02
  -6.9366946e-03  1.7383872e-02]
 [ 1.9689439e-02 -1.7978797e-02  2.3663705e-02 -1.6761661e-03
  -1.6853256e-02 -4.8478292e-03  3.1350451e-03 -3.0663736e-02
  -6.5563791e-03  1.8654943e-02]]
```

"Representing DNA sequences as Embedding Vectors. Convenient for machine learning, deep learning."

# Computer Aided Drug Design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2014.12.06



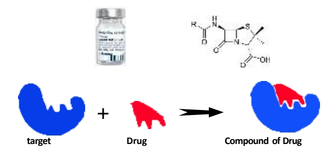
## OVerview

- Drus Design Process
- Overview of CADD (Computer-Aided Drug Design)



## The workings of Drugs.

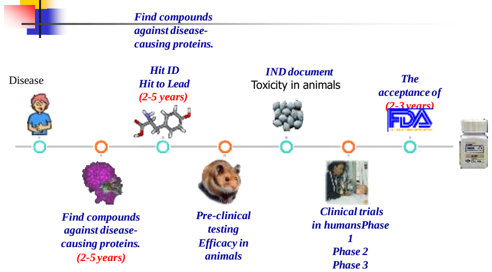
**Drug: A substance used in diagnosing, treating, or preventing a disease, or as a component of a drug.**




Lock-and-key model



## Drug development process



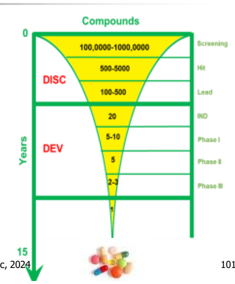



## TDrug development process

ou have to screen MANY compounds to find a drug

**10 – 15 years,**

**500 - 1000 million US $**






## Why we useCADD ?

*BrightFuture*

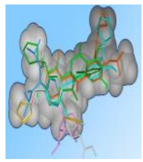
- ✓ Determine Hit
- ✓ Select leads
- ✓ Leading optimization

- Cost/time reduction
- Reduction in the number of animals used in experiments
- ncrease success rate
- Enhance understanding of drug-protein interactions

**Save 2 – 5 years, 100 - 500 million US $**

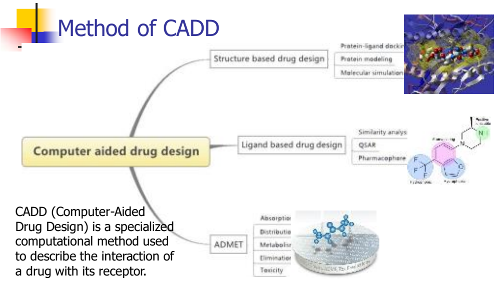
## What is Ligand ?

- Ligand is a molecule or ion capable of binding to another atom or molecule to form a complex. In the context of drugs and drug design, a ligand is often part of a drug molecule or another compound capable of interacting with a specific protein or tissue in the body.
- The interaction between a ligand and a protein can cause specific biological effects, such as inhibition or stimulation of a particular biological reaction. In ligand-based drug design, research focuses on identifying and optimizing ligands that effectively interact with the target protein to achieve therapeutic goals.





## Method of CADD




CADD (Computer-Aided Drug Design) is a specialized computational method used to describe the interaction of a drug with its receptor.



## Determine of 3D structure

- Principle: Using software to predict the 3D structure of a target protein based on the known 3D structure of a template protein.
- Tool: Modeller software.



## Similarity

- Principle: Starting from a lead compound, search for chemical compounds with similar structure or characteristics to the known compound.
- Methods: 3D and 2D similarity sequences.

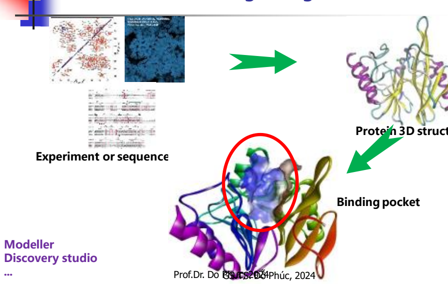


## Drug lead optimization

- After identifying a potential lead compound, optimize its structure.



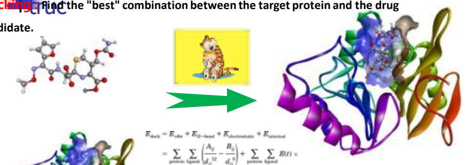
## Structure based drug design

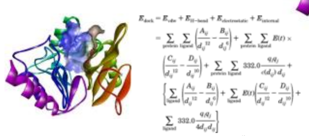



Modeller
Discovery studio
...

## Structure based Drug Designcấu trúc

**Docking**: **Find the "best" combination between the target protein and the drug candidate.**



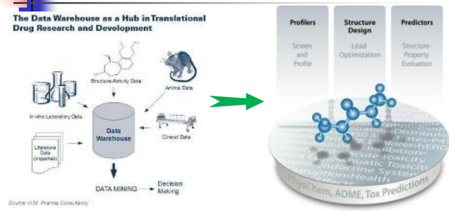


**Autodock**
**Glide**
**Sybyl**
**MOE**
**.....**



## ADMET eValuation

**Data mining....**




**Schrodinger**
**MOE**
**MedChem studio**
**.....**



## Conclusions

"With today's overwhelming data, computational methods have become essential in biological research.
Artificial intelligence (AI) helps tackle challenging biological problems based on vast biological data sources.
Some AI applications in biology include predicting secondary structures, Metagenomic Analysis, drug design,"....

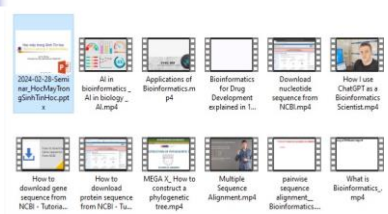


## Some of our papers in BioInformatics

- 1. Đỗ Phúc, Mai Xuân Hùng, Development of a hierarchical clustering algorithm, Journal of Science and Technology, Ho Chi Minh City University of Science, ISSN 1859-0218, Vol 3, pp 1-8, 2000
- 2. Hoang Kiem, Đỗ Phúc, Discovering motif-based association rules from sets of DNA sequences, Lecture Notes in Artificial Intelligence 2005, Springer, ISSN 0302-9743, RSCTC 2000, Banff, Canada, pp 386-390, 2000
- 3. Đỗ Phúc, Hoang Kiem, Developing a motif-based clustering algorithm to support similarity queries in a DNA sequence database, Journal of Science and Technology, VNU-HCM, ISSN 1859-0218, Vol 4+5, pp 83-89, 2001
- 4. Nguyễn Thanh Tùng, Trần Linh Thước, Đỗ Phúc, Prediction of protein secondary structure based on SCOP folds using Hidden Markov Model (HMM) and decision tree, Journal of Biotechnology, Vietnamese Academy of Science and Technology, ISSN: 1811-4989, Vol 2, Issue 4, pp 407-414, 2004
- 5. Đỗ Phúc, Hoàng Kiếm, Using the MTREE structure for similarity search in biological sequence databases, Biology, Vietnam Academy of Science and Technology, ISSN: 1811-4989, Vol 4, pp 151-158, 2006
- 6. Đỗ Phúc, Nguyễn Thị Kim Phụng, Using spectral vectors and M-Tree for graph clustering and searching in graph databases of protein structures, International Journal of Computer, Electrical, Automation, Control and Information Engineering, ISSN: 2010-3778, Vol 3, no. 8, 2008
- 7. Phạm Quốc Đàm, Đỗ Phúc, Lê Thị Thanh Mai, Enzyme classification prediction using data mining techniques, Journal of Science and Technology, Ho Chi Minh City University of Science, ISSN 1859-0218, Vol 11, issue 5, pp 247-252, 2008

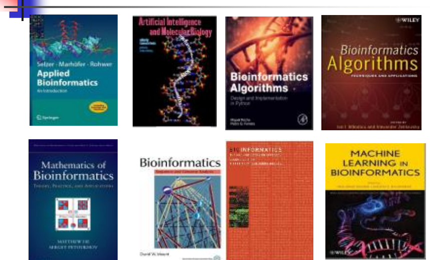


## Youtube Learning Materials





## References



## Future of bio informatics powered By AI




## Q&A

Thank you for your kind attention!